DAA
CSA0612

N. Thanveer Basha
192372122
CSE (AI)

9:- Brute Force Maximum Subarray

Given an array [-2, 3, 4, -1, -2, 1, 5, -3]

$A = [-2, 3, 4, -1, -2, 1, 5, -3]$

Steps:-

1. Select a starting index.
2. Index extend the subarray one element at a time.
3. Calculate the sum of each subarray.
4. Compare it with the current maximum sum.

An array size of $n = 8$

$$\frac{n(n+1)}{2} = \frac{8(9)}{2}$$

$$= \frac{\cancel{72}^{36}}{\cancel{2}_{1}}$$

$$= 36$$

Therefore, 36 subarrays are examined.

calculations of Different subarrays

Index 0

subarray	sum
[-2]	-2
[-2, 3]	1
[-2, 3, 4]	5
[-2, 3, 4, -1]	4
[-2, 3, 4, -1, -2]	2
[-2, 3, 4, -1, -2, 1]	3
[-2, 3, 4, -1, -2, 1, 5]	8
[-2, 3, 4, -1, -2, 1, 5, -3]	5

Index 1

subarray	sum
[3]	3
[3, 4]	7
[3, 4, -1]	6
[3, 4, -1, -2]	4

[3, 4, -1, -2, 1]	5
[3, 4, -1, -2, 1, 5]	10
[3, 4, -1, -2, 1, 5, -3]	7

Index 2

Subarray	Sum
[4]	4
[4, -1]	3
[4, -1, -2]	1
[4, -1, -2, 1]	2
[4, -1, -2, 1, 5]	7
[4, -1, -2, 1, 5, -3]	4

Index 3

Subarray	Sum
[-1]	-1
[-1, -2]	-3
[-1, -2, 1]	-2
[-1, -2, 1, 5]	3
[-1, -2, 1, 5, -3]	0

Index 4.

Subarray	sum
[-2]	-2
[-2,1]	-1
[-2,1,5]	4
[-2,1,5,-3]	1

Index 5

subarray	sum
[1]	1
[1,5]	6
[1,5,-3]	3

Index 6

subarray	sum
[5]	5
[5,-3]	2

Index 7

subarray	sum
[-3]	-3

Number of comparisons:-

36 subarrays is compared with the current maximum sum.

Maximum sum:-

The heighest sum obtained is

$$\boxed{10}$$

[3, 4, -1, -2, 1, 5]

verification

$$3+4-1-2+1+5=10$$

Time complexity:-

$$T(n) = O(n^2)$$

$$O(n^3)$$

Space complexity:-

(sum & Max sum)

$$\boxed{O(1)}$$

10:- Nearest Neighbor Distance (2D points)

Given points

A(1,2), B(4,6), C(7,8), D(2,3), E(5,5)

Distance formula :-

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

compute All pair Distances:-

pair	Distance
A - B	$\sqrt{25} = 5$
A - C	$\sqrt{72} = 8.49$
A - D	$\sqrt{2} = 1.41$
A - E	$\sqrt{25} = 5$
B - C	$\sqrt{13} = 3.61$
B - D	$\sqrt{13} = 3.61$
B - E	$\sqrt{2} = 1.41$
C - D	$\sqrt{50} = 7.07$
C - E	$\sqrt{13} = 3.61$
D - E	$\sqrt{13} = 3.61$

Num. of comparisons

$n = 5$ points

$$\frac{n(n-1)}{2} = \frac{5 \times 4}{2} = 10$$

Total comparisons = 10

Nearest pair

Minimum distance

$$\boxed{\sqrt{2} \approx 1.41}$$

A (1, 2) – D (2, 3)

B (4, 6) – E (5, 5)

Time Complexity:-

Best case $O(n^2)$

Average case $O(n^2)$

Worst case $O(n^2)$

Nearest pair A-D and B-E

Minimum Distance $\sqrt{2} \approx 1.41$

Total comparison 10

Time complexity $O(n^2)$